

Abdelrahman Elshrief

abdel.elshrief@mail.utoronto.ca | 613-770-4883 | linkedin.com/in/abdel-elshrief | github.com/Abdel-E | Website

EDUCATION

University of Toronto

Bachelor of Applied Science (BASc) in Computer Engineering

Toronto, ON

Expected: Apr 2029

- **Coursework:** Computer Fundamentals, Programming Fundamentals, Digital Systems, Signals and Systems

EXPERIENCE

Research Engineer Intern

May 2026 – Aug 2026

Royal Military College of Canada, NavINST Lab

Kingston, ON

- Enabled **multimodal perception and sensor fusion** for NavINST's autonomous vehicle by developing and calibrating **event-camera + frame-camera synchronization** for downstream computer vision pipelines.
- Enabled a **low-cost 4-channel GNSS array** by validating PlutoSDR calibration at **67 dB Signal-to-Noise Ratio** and evaluating **4 chirp/wideband calibration methods** for synchronized multi-channel SDR operation.
- Expanded GNSS interference test coverage across **3 jammer classes** by building a generator for **CW, chirp, and wideband** signals with configurable parameters and PSD/spectrogram diagnostics.
- Guided **multi-channel SDR calibration experiments** by authoring a technical literature review comparing chirp-based calibration methods, implementation tradeoffs, and experimental validation approaches.

Machine Learning Engineer Intern (Contract)

Jan 2026 – Apr 2026

Agentiiv

Toronto, ON

- Built a **Corporate Intelligence platform** for Agentiiv by architecting a **FastAPI MCP gateway** across **5+ integrations**.
- Reduced integration latency by **40%** while securing enterprise agent access with **JWT/RBAC** scoped authorization.
- Reduced deployment latency by **80%** by deploying Dockerized AWS microservices and building **Prometheus/Grafana** dashboards for service health and latency.

Technical Project Manager

Jan 2026 – Present

University of Toronto Machine Intelligence Team

Toronto, ON

- Leading **2 internal/external engineering projects**, driving delivery through Agile sprint planning, milestone tracking, and coordination across **teams of 6+ engineers**.

Autonomous Systems Developer

Sep 2025 – Apr 2026

University of Toronto Formula Racing Team

Toronto, ON

- Enabled **simulation-based perception testing** by programming **3+ virtual sensors**: LiDAR, IMU, and GPS, in **Unity/C#**.
- Improved **LiDAR ground detection by 10%** by applying **C++ linear algebra** to autonomous-vehicle sensor pipelines.

PROJECTS

Cerebras Kernel Challenge - Distributed Top-K kNN | CSL, Python, WSE-2, Parallel Computing

- Implemented **exact distributed Top-K kNN** across a 2D WSE-2 PE grid using query broadcast, vectorized L2 distance computation, and hierarchical Top-K reduction.
- Reduced kernel cycle counts by **up to ~85%**, cutting the baseline from **~160K to ~25K cycles** and K=256 from **~851K to ~318K** using vector FMACs, blocked memory layouts, and K-specific optimizations.

Twin - MakeUofT 1st Place Overall | Python, C++, Computer Vision, Raspberry Pi, Arduino

- Won **1st Place Overall** against **250+ hackers** by engineering a Pi/Arduino AI wearable and reducing vision inference latency by **80%** with localized YOLO-v8 models.

Telemetry Logs Risk Detector | Python, Scikit-learn, PyTorch, Pandas, NumPy

- Automated security-event triage by combining rule-based detection with **Isolation Forest** anomaly scoring to surface **exfiltration and anomalous behavior** through weighted risk alerts.

SKILLS

Languages: C, C++, Python, Java, Rust, Verilog, C#, SQL, TypeScript/JavaScript

Technologies: Node.js, React, PyTorch, Linux, Docker, AWS, CUDA, Computer Vision

Engineering: CI/CD, Git, Sensor Fusion, pytest, Agile/Scrum, Project Management, JIRA